

Technology Enabled Promotion of Indian Knowledge Traditions in Contemporary Education

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Abstract

Indian Knowledge System is the ancient India's vast wisdom spreading its branches along different domains including arts, science, philosophy, astronomy, music, dance, mathematics, architecture, yoga, medicine (Ayurveda) and education. It is passed from one generation to another having its roots in Vedas, Upanishads and ancient texts and various manuscripts. The National Education Policy (NEP 2020) besides transforming the traditional 10 + 2 education system to a 5 + 3 + 3 + 4 education pattern, highlights the importance of integration of traditional knowledge into modern education systems. In this context, the technology plays a crucial role and bridges the gap between the traditional knowledge system and modern education. This paper discusses how digitization, online learning platforms, mobile and web applications and emerging tools and technologies play a crucial role in making traditional education more accessible and interactive and integrating it with the contemporary education. This paper explores the various methods of preservation of ancient texts and manuscripts, making education more engaging and simpler, focusing on skill-based learning despite of grade-based learning, blending traditional education concepts with modern subjects, development of teacher along with students with the incorporation of various modern technologies such as Artificial Intelligence, Cloud Computing, Web and Mobile Applications, Augmented Reality and Virtual Reality. The paper also identifies the various challenges such as language barriers, cybersecurity issues and verification of authenticity of traditional knowledge. The paper concludes that by aligning technology with the goals of NEP 2020, the modern education system can effectively inherit the concepts from traditional knowledge system helping in a significant development of not only the students but also the teachers.

Keywords: Indian Knowledge Systems (IKS); Technology-Enabled Education; National Education Policy 2020 (NEP 2020); Digital Preservation of Traditional Knowledge; Emerging Educational Technologies

1) Introduction

Blend of ancient and diverse knowledge tradition makes Indian Knowledge System unique all across the world. From old scriptures like Vedas, Puranas and Epics, Indian Knowledge System has a wide range of subjects like mathematics, Yoga, Ayurveda, music, dance art and craft, etc. This tradition was passed from one generation to another by the system called Guru-Shishya Parampara. Although modern education provides skills based on current technology advancement, teachers can teach students in innovative ways which makes the session more interesting for students but it somewhere lacks to root the discipline a child develops in a value-based learning. Earlier students were sent to ashrams away from their home so that they can emphasize on ethical learning, practical understanding and overall development of mind and body. In current scenario, the education system is driven by the advancement of technology which is a totally different way of learning than the traditional educational system. In order to bridge the gap between the traditional and the modern education the New Education Policy (NEP 2020) was introduced. It basically emphasizes on skill-based learning so that the student knowledge along with discipline meets global standards. Technologies such as e-learning, Cloud Computing, Artificial Intelligence and Machine Learning offers innovative ideas for contemporary learners. Even with the help of technology the ancient manuscripts are accessible through digital repositories by which learners can easily access these materials from anytime anywhere.

This article offers an overview of a wide variety of technological methods that can complement existing educational practices in order to assist with the complete growth of all students. The research provides both an analytical and a descriptive perspective of various Indian knowledge traditions, as referenced in the National Education Policy 2020, and a number of technology-based education practices detailed in existing literature. The technologies analyzed are many different types of technologies that have the ability to create digital versions of manuscripts, use web and mobile applications, use artificial intelligence, employ cloud computing, employ augmented reality, employ virtual reality, and provide access to digitized Indian Knowledge through various means of preservation, access to information, and dissemination of that information. As part of the research, case studies have also been selected as examples of how these types of technology may assist in the real-world delivery of the Indian Knowledge Traditions. The overall focus of the research will be on the preservation of ancient texts, improvements to the teacher–learner relationship, increasing the qualifications of teachers, and

identifying the various challenges to be faced in this process, such as: language barriers, cybersecurity concerns, and verifying the authenticity of the content.

2) Background

2.1 Indian Knowledge Traditions: Conceptual Overview

The ancient Indian scriptures have been a rich source of information for exhaustive and intricate theories and practices called the Indian Knowledge Tradition, which includes several fields of study such as Philosophy, Mathematics, Art, Craft, Music, Dance, etc. The Indian Knowledge System has been designed to give students the ability to acquire the skills and knowledge required for professional success (in their respective fields) and to develop themselves into well-rounded, socially responsible members of society through the use of ethical values and spiritualism, self-discipline, and personal growth. Through the Indian Knowledge System, Students learn to think critically and make better decisions using the process of Guru-Shishya Parampara/Teacher-Student Relationship to establish a meaningful connection with their Gurus/Teachers. In addition to academics, the Indian Knowledge System supports Children's social, physical, mental, and creative growth through Holistic Development, as defined in the NEP 2020 framework. In order to succeed in today's increasingly globalized and digitalized world, a Multidisciplinary Approach must be implemented by integrating Traditional Methods of Learning, the Digital Marketplace, and Project-Based Learning.

2.2 New Education Policy 2020: Structural Shift from 10+2 to 5+3+3+4

One key suggestion of NEP 2020 is the depiction of the values, ethics and culture of India in contemporary curricula. In addition, technology will be a key component of improving the quality of education. An additional major shift from prior educational models was the introduction of a new curriculum model of a 5+3+3+4 structure to replace the former structure of 10+2, which identified three years for children in early childhood education, and two years for all other ages. In essence, the new curriculum model will facilitate early childhood care, foundational literacy, experiential learning and multi-disciplinary learning through the various years of a person's development. As such, it allows for the incorporation of ideas of the Indian

Knowledge System (such as storytelling, art, mathematics and ethics) from the initial stages of a child's development. The removal of the rigid boundaries between disciplines and providing students with flexibility in the selection of their stream of study is one of the highlights of NEP 2020 and is closely related to the character of the Indian Knowledge Traditions which encompasses science, philosophy, art and spirituality. Promoting the concept of holistic education, NEP 2020 encourages the integration of classical subjects (i.e. yoga, Ayurveda, Indian mathematics and Indian philosophy) with modern science and technology in the curricula.

3) Discussion

3.1 Integration of Indian Knowledge Systems under NEP 2020

The NEP 2020 document explicitly acknowledges that an embracing of the unique knowledge systems of India is essential to address many modern educational challenges. It identifies the numerous knowledge systems associated with many types of knowledge, such as classical texts, theories, traditional knowledge and indigenous practices, as holes that must be filled in developing curriculum, professional preparation for teachers and researchers, etc. Through introducing bilingual education with the use of regional language, NEP 2020 intends to make the knowledge associated with the knowledge systems more accessible and relevant to both local communities and the Global Village. By offering the school system with a shift from grading students to competency-based learning (skills) NEP 2020 aligns closely with traditional Indian schooling that has already focused on skill acquisition and application of knowledge in the real world through its knowledge systems. By introducing modern advancements in the educational processes, the Indian Knowledge Systems can be enhanced with practical skills, values, and cultural knowledge to create a complete, holistic and meaningful educational experience for students.

3.2 Role of Technology in promoting Indian Knowledge Traditions

Technology plays an essential role in preserving and connecting the traditional knowledge of ancient India with today's educational system. New technologies have enabled researchers, teachers, and students from all over the world to access rare manuscripts, palm leaves, and sacred texts through digital format in order to preserve these unique forms of knowledge while

making them available. Web and mobile apps have made knowledge of Indian disciplines like Yoga, Ayurveda, Philosophy, and Mathematics much more accessible and engaging for everyone via multimedia education, discussion forums, independent study opportunities, and increased availability. Emerging technologies such as Artificial Intelligence, Augmented Reality, and Virtual Reality provide unparalleled opportunities for many learners through immersive, hands-on experiences that help bring to life many aspects of Ancient Gurukuls, Astronomical Studies, and Architectural Wonders.

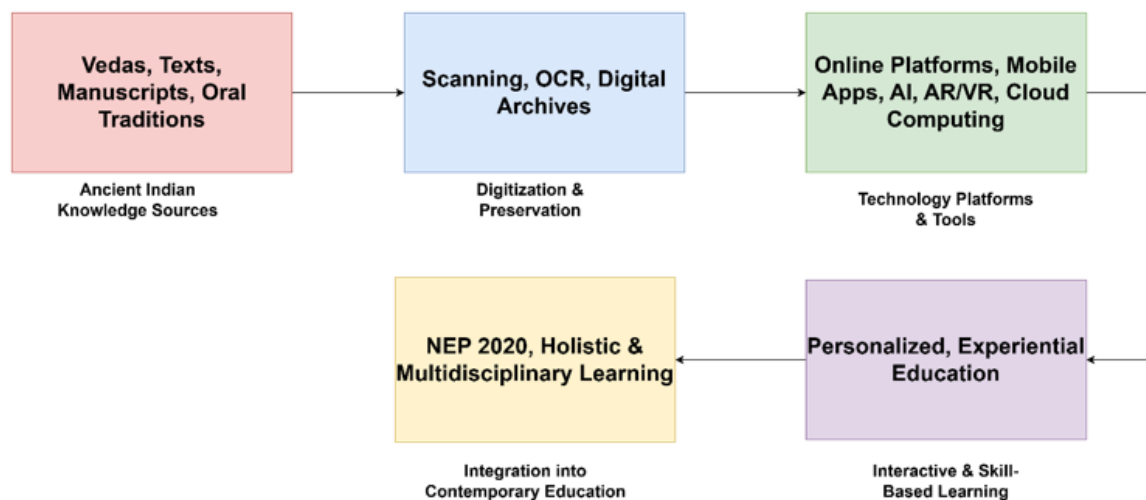


Figure 1: Technology-Enabled Promotion of Indian Knowledge Traditions

3.3 Emerging Technologies in Indian Knowledge Education

Technologies enable the development of an evolving Indian Knowledge education system using technology as transformative tools without repurposing educational philosophies developing a technology-enabled educational environment that is scalable, immersive, and driven by analytics, making Indian knowledge relevant in today's increasingly technological learning environment. The incorporation of AI has allowed Indian Knowledge to develop adaptive learning methodologies for the delivery of personalized content, provide the use of intelligent tutors, use translation assistant tools to translate from one language to another, and create adaptive learning environments that enhance the understanding of key concepts while maintaining independence. Augmented reality provides an interface through which users can visualize virtual information superimposed upon their real-world environment, which aids in expanding their knowledge of cultural and historical issues. Many of the concepts associated with augmented reality are related to architecture, ritual practices and astronomical

observations as well as the tools utilized by people of India in developing their systems of knowledge. Virtual Reality has made it possible to create immersive educational experiences that can replicate the learning environments, historical sites and events of the past. Experiential learning is enhanced through Virtual Reality learning, as well as cultural understanding and retention through this type of experiential learning. Using data analytics allows for monitoring and evaluating the results from these types of learning. Specifically, in IKS education, using data analytics allows for measuring learners' levels of understanding, the skills they possess, and to track learner engagement.

3.4 Technology-Enabled Teaching Learning

Education has undergone transformation into new definitions by introducing technology into traditional teaching-learning methods. In many ways, technology has moved away from the rigid content-based model of the past to a more flexible and learner-centered approach that allows teachers and students to have a greater role in determining the direction of education. Technology's application in the area of Indian Knowledge Systems provides an opportunity to provide a structure to the reconfiguration of ancient methods used to teach/learn by applying modern design practices. In this way, technology also enhances the ideals of traditional learning approaches (i.e. Holistic Development, Learning by Doing, and Value-based Education) within the current teaching environment.

3.5 Blended Learning Approaches with Experimental and Interactive Learning Methods

Blended learning is about finding a harmonious balance between face-to-face and online education. This has created a unique opportunity for educational institutions in India to provide their students with the benefits of both forms of education and training. The classroom setting facilitates learning through traditional learning theories, such as experience, theorizing, and discussing with an instructor. Online tools are an adjunct to help learners to further understand the materials through additional means of knowledge. The Indian Tradition of Knowledge focuses on learning through direct experiences, observational learning, and active participation; therefore, the use of technology to create an interactive learning environment; simulations, case studies, and problem-solving will increase the number of learners that benefit from traditional modes of instruction. Many traditional concepts of yoga, Ayurveda, Jyotish, and other skills have the potential to be delivered using digital technological methods, which allow for

conceptual visualization and knowledge application. Visualization of concepts aids in improving conceptual comprehension and helps eliminate the gap between the theoretical and practical aspects of the material.

3.6 Gamification of Indian Knowledge Content

Gamification is an approach to the learning process that provides the learner with a reward system as well as the game aspect. By adding these elements to the education process, individuals can have a positive experience while gaining knowledge about their culture's history, philosophy, and mathematics. Additionally, gamifying IKS will provide ways for presenting this type of content to learners in a fun, engaging way while maintaining the integrity of the knowledge presented. Furthermore, the use of e-Learning technology allows personalization in the educational process by providing learners with tailored content based on individual achievement, preference and ability. This personalization will be particularly beneficial for IKS learners, as these individuals typically have different cognitive styles, different linguistic backgrounds and varied levels of past experience. Using the advantages of technology, e-Learning allows learners to revisit complex subjects, explore areas of interest, and learn at a pace suitable for them. In this way, e-Learning provides an inclusive and differentiated learning opportunity.

3.7 Teacher Empowerment and Professional Development through Technology

Teachers play a vital role in interpreting and transmitting the knowledge systems of Indigenous Peoples in India. By using technology, teachers have greater access to digital repositories, access to continued professional training programs, access to technology-based collaboration tools across disciplinary boundaries, and access to new tools for innovating pedagogy. In addition, technology-supported continued professional development will provide teachers opportunities to bridge traditional scholarship with contemporary instructional practices to create greater academic rigor and enhance the effectiveness of their teaching.

3.8 Preservation and Authenticity of Indian Knowledge Traditions

The digital distribution of India's Knowledge Traditions raises questions of Preservation and Authenticity. Technology needs to be used to preserve and protect the Culture of India through

appropriate representation, utilization and sustainability. Today's advanced digitization technologies including high-resolution scanning, OCR and Metadata annotation allow access to Manuscripts for Academic use, while preserving them from deterioration. Recent advancements in digital technology enable us to preserve Ancient Manuscripts in digital format without causing deterioration of the manuscript physically. Digital Life Archives and repositories provide a means for organized storage, indexing and retrieval of resources for Indian Knowledge. Digital Life Archives provide a means for documenting scholarly citations for the documentation of Indian Knowledge, as well as for enabling organized distribution of Indian Knowledge. Authenticity must be validated by institutional means, through expert judgment by scholars, cross-textual analysis, and continued monitoring. Traditional scholars, scholars of various disciplines, and policymakers need to come together to establish a shared approach to preserving the intellectual integrity of Indian Knowledge from distortion. The ethical standards of digitizing respect Cultural Ownership, Intellectual Property rights, and Community Consent. The governance structure of Digital Life and Digital Archives needs to ensure that traditional Communities have a sense of dignity in regard to the treatment of their Traditional Knowledge, and are sensitive to the context of the knowledge they are preserving.

Table 1: Preservation Risks and Governance Strategies in Digital IKS Promotion

Risk Area	Potential Impact	Governance Strategy
Manuscript decay	Knowledge loss	Digital preservation
Content distortion	Misrepresentation	Expert validation
Unauthorized use	Cultural exploitation	Access control
Fragmentation	Loss of coherence	Centralized repositories
Ethical violations	Community distrust	Policy frameworks

4) Discussion

4.1 Case Study

This section of the paper represents various examples demonstrating how emerging technological developments/digital platforms enhance opportunities for people to be able to practice using technology to promote Indian Knowledge Traditions (IKTs) via contemporary educational environments. It highlights the ways in which traditional Indian knowledge is being preserved, distributed and incorporated within the modern education systems, according to the aims of NEP 2020.

4.1.1 The SWAYAM Platform – Promoting Indian Knowledge Systems

The SWAYAM (Study Webs of Active Learning for Young Aspiring Minds) Platform is a Massive Open Online Course (MOOC) initiative launched by the Government of India. SWAYAM has facilitated the democratization of education through its MOOC platform, SWAYAM, which provides Learners of all backgrounds access to the knowledge and skills found in many traditional Indian knowledge systems - Yoga, Philosophy, Sanskrit language, Ayurveda, Indian Mathematics, and Ethical Values; among many others - offered by prestigious institutions, including IITs, Central Universities, and National Institutes of Ayurveda.

Through video lecture presentations, Reading Materials, Forums, Self-Assessment Quizzes, and Certification Options on each Course, the SWAYAM platform provides the means by which to transfer the knowledge through a structured academic approach, to a wide audience of people coming from various geographic and socio-economic backgrounds.

The SWAYAM platform was designed as a Multilingual MOOC to create opportunities for people to use their own native languages, as recommended under the National Education Policy (NEP) 2020, which aims to make it easier for all people, regardless of location or location of their institution, to obtain an education.

Currently, SWAYAM provides the means to close the gap between traditional forms of Knowledge and Modern Methods of Education.

4.1.2 AR/VR (Augmented Reality and Virtual Reality) for Cultural and Historical Learning

Many startups and educational institutions in India are exploring the potential of AR and VR in recreating ancient Indian learning environments (Gurukuls), ancient universities such as Nalanda and Takshashila, as well as monuments like temples and observatories.

Immersive learning experiences allow students to experience the complete configuration of past educational environments, astronomical instruments, and methods of construction. The ability to learn in an interactive way helps students think about and understand history and culture better than they otherwise would. The use of AR/VR technologies in education supports the experiential and multidisciplinary approach stated in NEP 2020 and provides a strong means of preserving and revitalizing India's Knowledge Traditions.

4.1.3 National Digital Library of India (NDLI) for Preservation and Access

The NDLI (National Digital Library of India) is an extensive digital library that will collect, store, and distribute educational materials digitally. The very large collection of resources related to India's Knowledge System includes digitized manuscripts, classical works, valuable manuscripts, and research papers concerning India's knowledge system, culture, philosophy, and science. NDLI supports advanced searching, tagging of metadata, and supports multiple languages, so resources can be easily located and used for research or study. Long-term preservation of NDLI's materials will be performed through digital archives to protect ancient & rare documents from physical degradation. The institution also supports educators, researchers, and students by providing written materials that have been authenticated and verified for credibility and scholarly integrity concerning the Knowledge Traditions of India.

Table 2: Case Studies of Technology Enabled IKS Promotion

Case Study	Technology Used	IKS Domain	Educational Level	Key Outcomes
SWAYAM Platform	MOOC, LMS, Cloud	Philosophy, Yoga, Sanskrit	Higher Education	Wide accessibility, structured learning
NDLI	Digital Archiving, Metadata	Manuscripts, Texts	All levels	Preservation, authenticated access

AR/VR Applications	AR, VR	History, Architecture	School and Higher Education	Immersive learning
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4.2 Challenges in Technology-Based Promotion of Indian Knowledge

An array of structural and academic challenges faces the technology-facilitated dissemination of the Indian Knowledge Tradition that limits the successful application of such methods. One of the main difficulties, in connection with this, relates to how different languages and scripts have changed over time for the many texts of the Indian Knowledge Tradition, and thus require very specialized and in-depth academic understanding in order to accurately translate them and obtain their contextual meanings, thus resulting in inaccurate translations or watered-down meanings resulting from a lack of standardized digital resources which allow for translation of these texts. In addition, the existence of a "digital divide," i.e., variances in access to technology, access to the Internet, and/or digital literacy (which require significant investment of resources), prohibits the ability of all people to benefit from technology-enhanced learning on the Indian Knowledge Tradition. Moreover, determining the validity, reliability, and proper use of the digital form of Indian Knowledge is also difficult, given that many cyber platforms and their growing usage generates questions regarding the possible misuse of information that is circulated online, as well as possible duplication and monetization of elements of the Indian Knowledge Tradition without adequate validation. In addition, the existence of cyber threats and privacy concerns make it increasingly difficult to distribute materials, such as culturally sensitive Indian Knowledge, digitally. Educational innovation is constrained by the traditional view of education as an academic discipline as well as the reluctance to adopt modern technology in education. Meeting this challenge, and promoting an understanding of Indian Knowledge Traditions through the application of technology will take considerable planning and a tremendous amount of work by all those involved with Education.

Table 3: Impact of Technology-Enabled Teaching–Learning Models on Indian Knowledge Education

Dimension of Learning	Traditional Approach	Technology-Enabled Shift	Educational Impact
Knowledge Access	Limited to texts and teachers	Digital, anytime access	Wider reach
Learner Engagement	Passive participation	Interactive involvement	Higher motivation
Learning Pace	Uniform	Individualized	Inclusive education
Skill Development	Implicit	Explicit and measurable	Competency focus
Teacher Role	Instructor	Facilitator and mentor	Pedagogical innovation

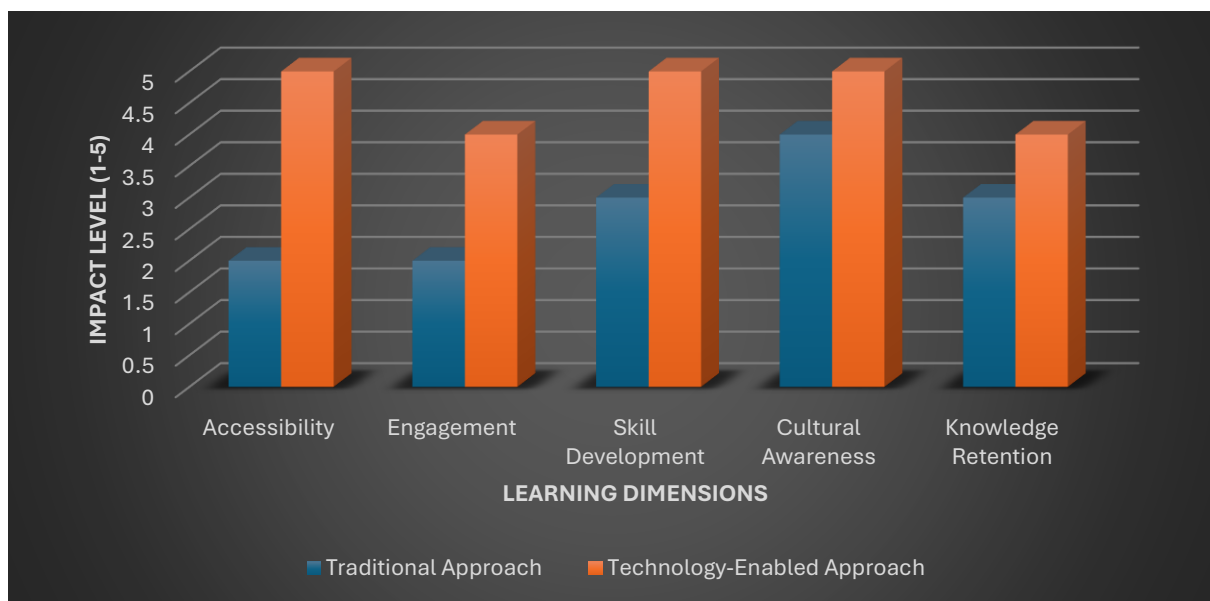


Figure 2: Conceptual Comparison of Traditional and Technology-Enabled Indian Knowledge Education

5) Conclusion

This study highlights how new technologies can assist in revitalizing and keeping alive the existing Indian Knowledge Systems (IKS). The benefit of combining new and immersive learning technologies with IKS has demonstrated how traditional methods of transmitting

knowledge can be modernized while still retaining their integrity. However, we cannot achieve this goal if we do not address critical issues such as differing languages, unequal access to technology, how we validate knowledge, and ethics surrounding the use of technology in education. To realize this opportunity, we need to build validation tools for IKS, promote multilingual learning platforms, and develop comprehensive learning technologies that empower learners. Technology-based IKS Education has the potential to transform education globally through innovative technology solutions.

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